

AI Applications Lecture 7

Embedding Layer and Distances/Similarities

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^aI thank my collaborator, Jing WANG, for cooperating in creating the teaching materials.

Outline

1. Introduction
2. Preliminaries: Mathematical Notations
3. How can a neural network represent the diversity of words? - Embedding
4. Can we use representations in a NN? - Distance and similarities
5. How can we input order information? - Positional encoding
6. Summary

Review of the previous lecture

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In this lecture, we reverse the perspective.

We will make explicit the fact that **when using a neural network as the core of a token generator, it implicitly converts discrete objects (tokens) into continuous objects (real-valued vectors)**.

The component responsible for this is the **embedding layer**.

Learning outcomes for this lecture

By the end of this lecture, you should be able to:

- Explain **what** an embedding layer does.

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By the end of this lecture, you should be able to:

- Explain **what** an embedding layer does.
- **Calculate** the **distances and similarities** (Euclidean distance, standard inner product, cosine similarity) between token representations (vectors) obtained through embedding.

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Preliminaries: mathematical notations

Set & Function:

- Sets: $\mathcal{A}$
- Membership: $x \in \mathcal{A}$
- Integer range: $[1, k]_{\mathbb{Z}} := \{1, \dots, k\}$
- Real numbers: $\mathbb{R}, \mathbb{R}_{>0}, \mathbb{R}_{\geq 0}$
- Function: $f : \mathcal{X} \rightarrow \mathcal{Y}$

Vector: Denoted by $\boldsymbol{v}$.

- A column of numbers, $\boldsymbol{v} \in \mathbb{R}^n$.
- i -th element is v_i .
- Standard inner product:
 $\langle \boldsymbol{u}, \boldsymbol{v} \rangle = \boldsymbol{u}^{\top} \boldsymbol{v}$

Preliminaries: mathematical notations

Sequence: Denoted by $\mathbf{a} = (a_1, a_2, \dots)$.

- A function $\mathbf{a} : [1, n]_{\mathbb{Z}} \rightarrow \mathcal{A}$.
- Length is denoted by $|\mathbf{a}|$.

Matrix: Denoted by $\mathbf{A}$.

- $m \times n$ matrix: $\mathbf{A} \in \mathbb{R}^{m,n}$.
- (i, j) -th element is $a_{i,j}$.

Tensor: Denoted by $\underline{\mathbf{A}}$.

- Simply a multi-dimensional array.
- Vector $\rightarrow$ 1st-order, Matrix $\rightarrow$ 2nd-order.

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Question: Words are not uniform. The one-hot encoding doesn't seem to represent such non-uniformity. How

Question: Words are not uniform. The one-hot encoding doesn't seem to represent such non-uniformity. How can the neural network represent each word's characteristic?

Symmetry and fully-connected layer

When the **input is one-hot**, we have no choice but to treat **each input node equally**. The natural connection to the next set of nodes is a **fully-connected** layer.

Symmetry and fully-connected layer

When the **input is one-hot**, we have no choice but to treat **each input node equally**. The natural connection to the next set of nodes is a **fully-connected** layer.

This part of the network is called an **embedding layer**.

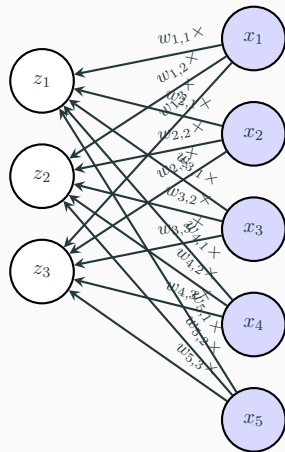


Figure 1: A fully-connected layer.

Matrix representation and equivalence to linear regression

A fully-connected layer is a **linear map** and can be written using a matrix

$\mathbf{W} \in \mathbb{R}^{d_{\text{emb}}, d_{\text{in}}}$:

$$\mathbf{z} := \mathbf{W} \mathbf{x} \quad (\mathbf{x} \in \mathbb{R}^{d_{\text{in}}}, \mathbf{z} \in \mathbb{R}^{d_{\text{emb}}}). \quad (1)$$

Active edges for a one-hot input

When the input is a one-hot vector,
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When the input is a one-hot vector,
 $x = e_i$, something interesting happens.
The multiplication $W e_i$ simply selects
the i -th **column of the matrix W** .

$$W e_i = w_i \quad (2)$$

(where w_i is the i -th column of W).

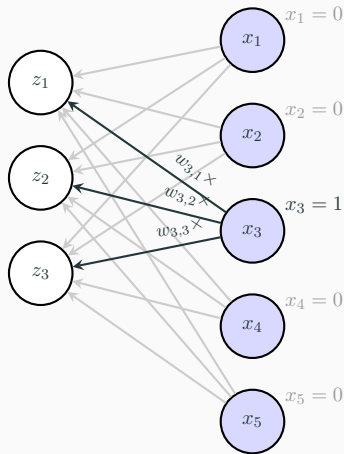


Figure 2: Active edges for input e_3 .

Equivalent understanding of embedding

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- "Feeding a one-hot vector e_i into the first fully-connected layer"...

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- "Feeding a one-hot vector e_i into the first fully-connected layer"...

...is **equivalent** to...

- "Directly looking up the column vector w_i in the weight matrix".

The column vector w_i is a **learnable parameter** vector that represents token i .

Equivalent understanding of embedding

Definition (Embedding)

The map defined by each column $\boldsymbol{w}_i$ of the matrix $\boldsymbol{W} \in \mathbb{R}^{d_{\text{emb}}, d_{\text{in}}}$,

$$\iota : i \in \{1, \dots, d_{\text{in}}\} \mapsto \boldsymbol{w}_i \in \mathbb{R}^{d_{\text{emb}}} \quad (3)$$

is called an **embedding** or **embedding representation** [1, 2, 3].

Equivalent understanding of embedding

Remark

It is customary to call w_i the **representation** of the i -th token or the **embedding** of the i -th token.

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Mathematically, an "embedding" is an **injective map** that preserves structure.

Equivalent understanding of embedding

Remark

It is customary to call w_i the **representation** of the i -th token or the **embedding** of the i -th token.

Mathematically, an "embedding" is an **injective map** that preserves structure.

This name is used with the expectation that the relationships between tokens will be **reflected** in the relationships between their vector representations in the new space.

Vocabulary extension (adding columns)

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How do we add new tokens to our vocabulary?

- We increase the dimension of the one-hot vectors (i.e., increase d_{in}).
- In the matrix representation, we simply **add new column vectors to W** .
- It is possible to **train only the new columns**, or fine-tune the entire matrix.

Embedding - answer

Question: How can the neural network represent each word's characteristic?

Answer:

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Answer: The weights of the edges connecting to the input nodes implicitly represent the characteristics of each token.

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Distances and similarities in the embedding space

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If the embedding reflects semantic information, calculating the **relationships between vectors** can have practical applications [2, 1].

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Here, we will strictly define the most fundamental measures.

Distances and similarities in the embedding space

Definition (Euclidean Distance)

For $\mathbf{u}, \mathbf{v} \in \mathbb{R}^{d_{\text{emb}}}$, the **Euclidean distance** $d_{\text{E}}(\mathbf{u}, \mathbf{v})$ is defined as:

$$d_{\text{E}}(\mathbf{u}, \mathbf{v}) := \|\mathbf{u} - \mathbf{v}\|_2 = \sqrt{(\mathbf{u} - \mathbf{v})^\top (\mathbf{u} - \mathbf{v})}. \quad (4)$$

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The **smaller** the distance, the more similar the concepts.

Distances and similarities in the embedding space

Definition (Standard Inner Product)

For $\mathbf{u}, \mathbf{v} \in \mathbb{R}^{d_{\text{emb}}}$, the **Standard Inner Product** $\langle \mathbf{u}, \mathbf{v} \rangle$ is defined as:

$$\langle \mathbf{u}, \mathbf{v} \rangle := \mathbf{u}^\top \mathbf{v} = \sum_{k=1}^{d_{\text{emb}}} u_k v_k. \quad (5)$$

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$$\langle \mathbf{u}, \mathbf{v} \rangle := \mathbf{u}^\top \mathbf{v} = \sum_{k=1}^{d_{\text{emb}}} u_k v_k. \quad (5)$$

The **larger** the inner product, the more similar the concepts. The Transformer architecture [5] uses this measure extensively.

Distances and similarities in the embedding space

Definition (Cosine Similarity)

For $\mathbf{u}, \mathbf{v} \in \mathbb{R}^{d_{\text{emb}}} \setminus \{0\}$, the **cosine similarity** $\text{cos}(\mathbf{u}, \mathbf{v})$ is defined as:

$$\text{cos}(\mathbf{u}, \mathbf{v}) := \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{u}\|_2 \|\mathbf{v}\|_2}. \quad (6)$$

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This is the cosine of the angle between the two vectors. The **larger** the similarity (closer to 1), the more similar the concepts.

Distances and similarities in the embedding space

The following identity holds:

$$\begin{aligned}\|\mathbf{u} - \mathbf{v}\|_2^2 &= \|\mathbf{u}\|_2^2 + \|\mathbf{v}\|_2^2 - 2 \langle \mathbf{u}, \mathbf{v} \rangle \quad (7) \\ &= \|\mathbf{u}\|_2^2 + \|\mathbf{v}\|_2^2 \\ &\quad - 2 \|\mathbf{u}\|_2 \|\mathbf{v}\|_2 \cos(\mathbf{u}, \mathbf{v})\end{aligned}$$

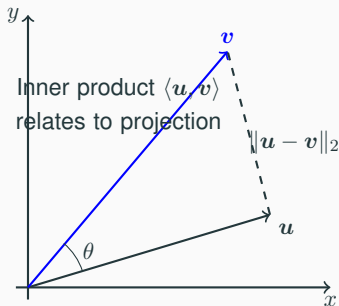


Figure 3: Geometry of measures.

Examples and exercises

Example (Euclidean Distance)

For $u = \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}$ and $v = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$, find the Euclidean distance.

Examples and exercises

Step 1: Calculate the difference vector

$$u - v = \begin{bmatrix} 2 - 1 \\ -1 - 2 \\ 3 - (-1) \end{bmatrix} = \begin{bmatrix} 1 \\ -3 \\ 4 \end{bmatrix} \quad (8)$$

Examples and exercises

Step 2: Calculate the L2 norm of the difference

$$\|u - v\|_2 = \sqrt{1^2 + (-3)^2 + 4^2} \quad (9)$$

$$= \sqrt{1 + 9 + 16} = \sqrt{26}. \quad (10)$$

The Euclidean distance is $\sqrt{26}$.

Examples and exercises

Exercise

Find the Euclidean distance $d_E(\mathbf{a}, \mathbf{b})$ for $\mathbf{a} = \begin{bmatrix} -1 \\ 4 \\ 2 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} 3 \\ 0 \\ -2 \end{bmatrix}$.

Examples and exercises

Answer

The difference vector is:

$$\mathbf{a} - \mathbf{b} = \begin{bmatrix} -1 - 3 \\ 4 - 0 \\ 2 - (-2) \end{bmatrix} = \begin{bmatrix} -4 \\ 4 \\ 4 \end{bmatrix} \quad (11)$$

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The Euclidean distance is:

$$\|\mathbf{a} - \mathbf{b}\|_2 = \sqrt{(-4)^2 + 4^2 + 4^2} = \sqrt{16 + 16 + 16} = \sqrt{48} = 4\sqrt{3}. \quad (12)$$

Examples and exercises

Example (Inner Product)

For $u = \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}$ and $v = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$, find the inner product.

Examples and exercises

$$\langle u, v \rangle = 2 \cdot 1 + (-1) \cdot 2 + 3 \cdot (-1) \quad (13)$$

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$$\langle \mathbf{u}, \mathbf{v} \rangle = 2 \cdot 1 + (-1) \cdot 2 + 3 \cdot (-1) \quad (13)$$

$$= 2 - 2 - 3 = -3. \quad (14)$$

The inner product is -3 .

Examples and exercises

Exercise

Find the inner product $\langle a, b \rangle$ for $a = \begin{bmatrix} -1 \\ 4 \\ 2 \end{bmatrix}$ and $b = \begin{bmatrix} 3 \\ 0 \\ -2 \end{bmatrix}$.

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Answer

$$\langle a, b \rangle = (-1) \cdot 3 + 4 \cdot 0 + 2 \cdot (-2) \quad (15)$$

Examples and exercises

Answer

$$\langle a, b \rangle = (-1) \cdot 3 + 4 \cdot 0 + 2 \cdot (-2) \quad (15)$$

$$= -3 + 0 - 4 = -7. \quad (16)$$

Examples and exercises

Example (Cosine Similarity)

For $u = \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}$ and $v = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$, find the cosine similarity.

Examples and exercises

Step 1: Reuse the inner product

$$\langle u, v \rangle = -3 \quad (\text{from previous example}) \quad (17)$$

Examples and exercises

Step 2: Calculate the norms

$$\|\mathbf{u}\|_2 = \sqrt{2^2 + (-1)^2 + 3^2} = \sqrt{4 + 1 + 9} = \sqrt{14} \quad (18)$$

$$\|\mathbf{v}\|_2 = \sqrt{1^2 + 2^2 + (-1)^2} = \sqrt{1 + 4 + 1} = \sqrt{6} \quad (19)$$

Examples and exercises

Step 3: Divide inner product by product of norms

$$\cos(\mathbf{u}, \mathbf{v}) = \frac{-3}{\sqrt{14} \sqrt{6}} = \frac{-3}{\sqrt{84}} = \frac{-3}{2\sqrt{21}}. \quad (20)$$

Examples and exercises

Exercise

Find the cosine similarity for $a = \begin{bmatrix} -1 \\ 4 \\ 2 \end{bmatrix}$ and $b = \begin{bmatrix} 3 \\ 0 \\ -2 \end{bmatrix}$.

Examples and exercises

Answer

$$\langle \mathbf{a}, \mathbf{b} \rangle = -7 \quad (\text{from previous exercise}) \quad (21)$$

$$\|\mathbf{a}\|_2 = \sqrt{(-1)^2 + 4^2 + 2^2} = \sqrt{1 + 16 + 4} = \sqrt{21} \quad (22)$$

$$\|\mathbf{b}\|_2 = \sqrt{3^2 + 0^2 + (-2)^2} = \sqrt{9 + 0 + 4} = \sqrt{13} \quad (23)$$

$$\cos(\mathbf{a}, \mathbf{b}) = \frac{-7}{\sqrt{21} \sqrt{13}} = \frac{-7}{\sqrt{273}}. \quad (24)$$

Distance and similarity - answer

Question: An embedding layer in a NN convert each token to a vector. Can we interpret them?

Answer: We can quantify the distance or similarity of token representation vectors in the NN.

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Question: The meaning of a token can depend strongly on its position in a sequence. How can we input **order information** into a Transformer?

Why do we need positional information?

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This has important advantages:

- The same parameters can be reused at every position, giving good **parameter efficiency**.
- The same architecture can be applied directly to **variable-length sequences**.

However, this symmetry also creates a problem: without positional information, the network does not intrinsically know **where** each token appears.

Permutation equivariance without positional information

Let

$$t = \left(t^{(1)}, t^{(2)}, \dots, t^{(L)}\right) \tag{25}$$

be an input sequence, and let σ be a permutation of its positions.

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A Transformer constructed only from position-wise operations and self-attention, with **no positional information and no position-dependent mask**, is permutation equivariant:

$$f_{\theta}(\sigma(\mathbf{t})) = \sigma(f_{\theta}(\mathbf{t})) . \quad (26)$$

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Thus, permuting the input tokens merely permutes the corresponding outputs in the same way.

Why is permutation equivariance a problem?

Consider the two sequences

(wine, red) and (red, wine). (27)

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If the network has no information about token positions, exchanging the two tokens can only exchange their corresponding output representations.

However, in natural language, **changing word order can substantially change the meaning.**

Therefore, token identity alone is not sufficient. We also want to tell the network **where each token occurs.**

Adding position to the input

Instead of regarding the input only as

$$\mathbf{t} = \left(t^{(1)}, t^{(2)}, \dots, t^{(L)} \right), \quad (28)$$

we regard each token together with its position:

$$\left((1, t^{(1)}), (2, t^{(2)}), \dots, (L, t^{(L)}) \right). \quad (29)$$

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- **what** the token is, and
- **where** the token appears.

Mechanisms for supplying such positional information are called **positional encodings**.

Positional embedding

Definition (Positional Embedding)

Let the possible positions be $[1, L]_{\mathbb{Z}}$. A vector-valued map

$$\mathbf{p} : i \in [1, L]_{\mathbb{Z}} \longmapsto \mathbf{p}_i \in \mathbb{R}^{d_{\text{pos}}} \quad (30)$$

that converts the position i into a vector representation used by a neural network is called a **positional embedding**.

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Thus, a positional embedding represents

$$\text{position} \longrightarrow \text{vector}. \quad (31)$$

Token embedding and positional embedding

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If $\mathbf{p}_i \in \mathbb{R}^{d_{\text{emb}}}$ one simple way to provide both pieces of information is

$$\mathbf{x}^{(i)} = \iota(t^{(i)}) + \mathbf{p}_i. \tag{32}$$

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If $p_i \in \mathbb{R}^{d_{\text{emb}}}$ one simple way to provide both pieces of information is

$$\mathbf{x}^{(i)} = \iota(t^{(i)}) + p_i. \quad (32)$$

The representation $\mathbf{x}^{(i)}$ therefore depends on both the **token** $t^{(i)}$ and its **position** i .

Example 1: sinusoidal positional embedding

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Assume that d_{emb} is even. For

$$k \in \left[0, \frac{d_{\text{emb}}}{2} - 1\right]_{\mathbb{Z}}, \quad (33)$$

define

$$(\mathbf{p}_i)_{2k+1} := \sin \left(\frac{i}{10000^{2k/d_{\text{emb}}}} \right), \quad (34)$$

$$(\mathbf{p}_i)_{2k+2} := \cos \left(\frac{i}{10000^{2k/d_{\text{emb}}}} \right). \quad (35)$$

Sinusoidal positional embedding

The positional embedding is therefore the function

$$i \longmapsto \mathbf{p}_i = \begin{bmatrix} \sin(i) \\ \cos(i) \\ \sin(i/10000^{2/d_{\text{emb}}}) \\ \cos(i/10000^{2/d_{\text{emb}}}) \\ \vdots \end{bmatrix} \in \mathbb{R}^{d_{\text{emb}}}. \quad (36)$$

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Different coordinates oscillate at different frequencies.

The original Transformer supplies position information by adding this vector to the token embedding:

$$\mathbf{x}^{(i)} = \iota(t^{(i)}) + \mathbf{p}_i. \quad (37)$$

Example 2: Rotary Position Embedding (RoPE)

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Another widely used positional encoding is the **Rotary Position Embedding (RoPE)** [4].

RoPE maps a position to a **rotation matrix**: $i \longmapsto R_i$.

Thus, under the vector-valued definition given above, RoPE is more precisely a **positional encoding mechanism** rather than an ordinary positional embedding vector.

Two-dimensional rotation

First define the two-dimensional rotation matrix

$$\text{Rot}(\phi) := \begin{bmatrix} \cos \phi & -\sin \phi \\ \sin \phi & \cos \phi \end{bmatrix}. \quad (38)$$

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Consequently,

$$\text{Rot}(\phi)^T \text{Rot}(\psi) = \text{Rot}(\psi - \phi). \quad (41)$$

RoPE as a function from position to a matrix

Let the dimension be $d_{\text{rot}} = 2m$ and choose frequencies

$$\theta_k := 10000^{-2(k-1)/d_{\text{rot}}} \quad (k \in [1, m]_{\mathbb{Z}}). \quad (42)$$

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$$\mathbf{R}_i := \begin{bmatrix} \text{Rot}(i\theta_1) & & \mathbf{0} \\ & \ddots & \\ \mathbf{0} & & \text{Rot}(i\theta_m) \end{bmatrix} \in \mathbb{R}^{d_{\text{rot}}, d_{\text{rot}}}. \quad (43)$$

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Hence RoPE can be regarded as the function

$$i \longmapsto \mathbf{R}_i. \quad (44)$$

RoPE converts absolute positions into relative positions

Consider arbitrary vectors $\mathbf{v}^{(i)}, \mathbf{v}^{(j)} \in \mathbb{R}^{d_{\text{rot}}}$, corresponding to position i and j .

Their RoPE-transformed inner product is

$$(\mathbf{R}_i \mathbf{v}^{(i)})^\top (\mathbf{R}_j \mathbf{v}^{(j)}) = (\mathbf{v}^{(i)})^\top \mathbf{R}_i^\top \mathbf{R}_j \mathbf{v}^{(j)} = (\mathbf{v}^{(i)})^\top \mathbf{R}_{j-i} \mathbf{v}^{(j)}. \quad (45)$$

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Thus, $(\mathbf{R}_i \mathbf{u})^\top (\mathbf{R}_j \mathbf{v})$ depends on the positions i and j only through their **relative displacement** $j - i$.

RoPE converts absolute positions into relative positions

For each two-dimensional block,

$$\text{Rot}(i\theta_k)^\top \text{Rot}(j\theta_k) = \text{Rot}((j - i)\theta_k). \quad (46)$$

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Therefore,

$$\mathbf{R}_i^\top \mathbf{R}_j = \begin{bmatrix} \text{Rot}((j-i)\theta_1) & & \mathbf{0} \\ & \ddots & \\ \mathbf{0} & & \text{Rot}((j-i)\theta_m) \end{bmatrix} = \mathbf{R}_{j-i}. \quad (47)$$

Positional encoding - answer

Question: How can we input order information into a Transformer?

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Answer: We append the position index to each token.

Outline

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2. Preliminaries: Mathematical Notations
3. How can a neural network represent the diversity of words? - Embedding
4. Can we use representations in a NN? - Distance and similarities
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6. Summary

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Summary

Let's summarize the key takeaways from today's lecture.

- An **embedding layer** is a fully-connected layer that takes a one-hot vector as input. This is equivalent to a lookup operation, where the i -th token is mapped to the i -th column vector w_i of the layer's weight matrix.

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- This vector w_i is called the **representation** or **embedding** of token i . It's a dense, continuous vector in a lower-dimensional space.
- We defined three key measures to quantify relationships between embeddings: **Euclidean distance**, **standard inner product**, and **cosine similarity**. These allow us to capture the notion of "similarity" in the embedding space.

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Attention is all you need.

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